

image of at most $k!$ injective homomorphisms $H \rightarrow G$, so

$$h(G) \leq k! \cdot d \|G\| \leq k! \cdot d \cdot n^2 \leq \eta n_0 \cdot n^2 \leq \eta n^3.$$

The removal lemma therefore yields a set E of at most ηn^2 edges in G such that $H \not\subseteq G - E$. Like every edge of G , each of the edges in E lies in at most d copies of H in G . So we also have

$$h(G) \leq k! \cdot d |E|.$$

As the copies of H are thinly spread over G , every edge of G lies in the image of some injective homomorphism $H \rightarrow G$. Hence $h(G) \geq \|G\|/m$, so

$$\|G\| \leq m h(G) \leq m k! d |E| \leq m k! d \eta n^2 = \delta n^2.$$

As the number h of copies of H in G is no bigger than $h(G)$, the above inequality also shows $h \leq h(G) \leq \delta n^2$, as claimed. $\square$

[7.7.3] **Corollary 7.6.5.** *For every $\delta > 0$ there exists an integer n_0 such that the following holds for every graph G of order $n \geq n_0$: if every edge of G lies in exactly one triangle, then G has at most δn^2 edges.* $\square$

7.7 Szemerédi's theorem

When Szemerédi proved his first version of the regularity lemma, he did not think of it as a precursor to the fundamental structure theorem for all of extremal graph theory that it has since become. He thought of it, well, as a lemma: a lemma he needed in a proof of a conjecture of Erdős and Turán about ‘arithmetic progressions’ of integers. We shall describe this theorem in this section, and then use the tools from Section 7.6 to derive its simplest case in just over a page.

arithmetic
progression

length An *arithmetic progression* is an increasing sequence of integers of the form $a, a + d, a + 2d, \dots$, one in which adjacent terms always differ by the same amount. The number of terms in the sequence is its *length*. One of the earliest results in combinatorics is *van der Waerden's theorem* that, no matter how we partition the natural numbers into finitely many classes, one of these will contain arbitrarily long arithmetic progressions.¹⁸ Szemerédi's theorem explores the deeper reasons why this is so.

upper
density

In any partition $\{A_1, \dots, A_p\}$ of $\mathbb{N}$ into finitely many classes, one of these A_i will have positive *upper density*: for some $\delta > 0$, in our case certainly for $\delta = 1/p$, arbitrarily large $n \in \mathbb{N}$ will satisfy $|A_i \cap n| \geq \delta n$

¹⁸ Compare Exercise 5 in Chapter 9.

(where $n = \{0, \dots, n-1\}$). Szemerédi's theorem says that the latter is enough to assume to get arbitrarily long arithmetic progressions in A_i . We do not need a choice between A_i and other partition sets, as in van der Waerden's theorem, we just need a single set that is dense enough:

Theorem 7.7.1. (Szemerédi 1975)

Every set $A \subseteq \mathbb{N}$ of positive upper density contains arbitrarily long arithmetic progressions.

Szemerédi's theorem is often expressed in more compact notation. For all $k, n \in \mathbb{N}$, and abbreviating 'arithmetic progression' as 'AP', let

AP

$$r_k(n) := \max \{ |D| : D \subseteq n \text{ and } D \text{ contains no AP of length } k \}.$$

Thus, if $k \in \mathbb{N}$ is such that $\lim_{n \rightarrow \infty} r_k(n)/n = 0$ then, for any sets $D_n \subseteq n$ that contain no arithmetic progression of length k , the values of $|D_n|/n$ will tend to zero as n grows. But if $A \subseteq \mathbb{N}$ has positive upper density, the values of $|A \cap n|/n$ exceed some $\delta > 0$ again and again. For those n , the sets $A \cap n$ will eventually contain arithmetic progressions of length k . This happens for every k , just for later n when k is bigger. Hence the assertions that $\lim_{n \rightarrow \infty} r_k(n)/n = 0$, one for each k , together imply Theorem 7.7.1.

For this reason, Szemerédi's theorem is often stated as follows:

Theorem 7.7.2. *For all integers $k \geq 3$ we have $\lim_{n \rightarrow \infty} r_k(n)/n = 0$.*

Even given the regularity lemma, Szemerédi's theorem is still a deep result far beyond the scope of this book. Indeed, it is surprising that an arithmetic statement such as this is susceptible to a purely combinatorial proof at all. The case of $k = 3$, however, which is known as Roth's theorem (1953), follows easily from Corollary 7.6.5:

Proposition 7.7.3. $\lim_{n \rightarrow \infty} r_3(n)/n = 0$.

Proof. We show that, for every $\epsilon > 0$, we have $r_3(n)/n \leq \epsilon$ for every n greater than the n_0 returned by Corollary 7.6.5 on input $\delta := \epsilon/12$. To prove this for given $n \geq n_0$, consider any set $D \subseteq n$ that contains no arithmetic progression of length 3. We have to show that $|D| \leq \epsilon n$.

[7.6.5]

 ϵ, δ, n_0 D

We shall apply Corollary 7.6.5 to the tripartite graph G with vertex classes $X := \{1, \dots, n\}$ and $Y := \{1, \dots, 2n\}$ and $Z := \{1, \dots, 3n\}$, viewed as disjoint sets. As the edges of G we take those of its *defining triangles*, the complete graphs $K(x, d) = K^3$ with vertices

 X, Y, Z

defining

triangle

 $K(x, d)$

$$x \in X \quad \text{and} \quad x + d \in Y \quad \text{and} \quad 2x + d \in Z$$

 $E(G)$

for all $x \in X$ and $d \in D$.

Since any two vertices of a defining triangle determine its third vertex, the defining triangles are edge-disjoint. Thus, every edge of G lies in a unique defining triangle, and

$$|G| = 6n \quad \text{and} \quad \|G\| = 3n|D|.$$

Let us show that G has no other triangles than the defining ones, and therefore satisfies the premise of Corollary 7.6.5.

Suppose $T \subseteq G$ is a non-defining triangle. Let T_1, T_2, T_3 be the defining triangles that contributed its edges. They are distinct, because any triangle that contains two of the edges of T contains all its vertices, and hence equals T . Since there are no edges inside the partition classes X, Y, Z , our T has one vertex in each class (Fig. 7.7.1, left).

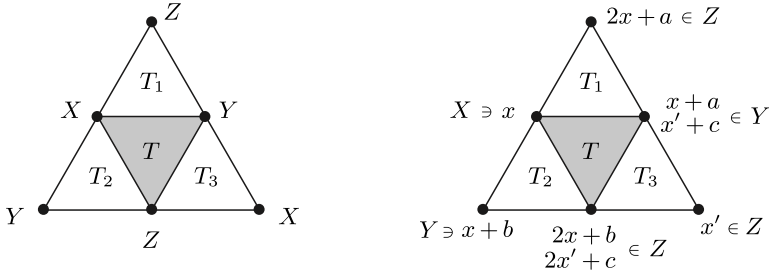


Fig. 7.7.1. Three defining triangles creating another triangle

Assume that the X - Y edge of T was contributed by T_1 , and its X - Z edge by T_2 . The vertex x of T in X then lies in $T_1 \cap T_2$, so

$$T_1 = K(x, a) \quad \text{and} \quad T_2 = K(x, b) \quad \text{and} \quad T_3 = K(x', c)$$

for some $a, b, c \in D$ and $x' \in X$. Since T_3 contributes the Y - Z edge of T but not its other edges, we have $x' \neq x$. The vertices of T , then, are

$$x \in X \quad \text{and} \quad x + a = x' + c \in Y \quad \text{and} \quad 2x + b = 2x' + c \in Z$$

(Fig. 7.7.1, right), so

$$a = c + (x' - x) \quad \text{and} \quad b = c + 2(x' - x).$$

Hence either (c, a, b) or (b, a, c) is an arithmetic progression of length 3 in D , with difference $|x' - x| \neq 0$ since $x' \neq x$. This contradicts the choice of D , so we conclude that G contains no triangles other than the defining ones. In particular, every edge of G lies in exactly one triangle.

Corollary 7.6.5 now yields

$$\|G\| \leq \delta \cdot |G|^2 = \delta \cdot (6n)^2.$$

As $\|G\| = 3n|D|$, we conclude that

$$|D| = \|G\|/3n \leq \delta \cdot 12n = \epsilon n$$

as desired. $\square$

Exercises

- 1.⁻ Show that $K_{1,3}$ is extremal without a P^3 .
- 2.⁻ Given $k > 0$, determine the extremal graphs of chromatic number at most k .
- 3.⁻ Is there a graph that is edge-maximal without a K^3 minor but not extremal?
4. Determine the value of $\text{ex}(n, K_{1,r})$ for all $r, n \in \mathbb{N}$.
- 5.⁺ Given $k > 0$, determine the extremal graphs without a matching of size k .
(Hint. Theorem 2.2.3 and Ex. 20, Ch. 2.)
6. Without using Turán's theorem, show that the maximum number of edges in a triangle-free graph of order $n > 1$ is $\lfloor n^2/4 \rfloor$.
7. Show that

$$t_{r-1}(n) \leq \frac{1}{2}n^2 \frac{r-2}{r-1},$$

with equality whenever $r-1$ divides n .

8. Show that $t_{r-1}(n)/\binom{n}{2}$ converges to $(r-2)/(r-1)$ as $n \rightarrow \infty$.
(Hint. $t_{r-1}((r-1)\lfloor \frac{n}{r-1} \rfloor) \leq t_{r-1}(n) \leq t_{r-1}((r-1)\lceil \frac{n}{r-1} \rceil)$.)
9. Does every large enough graph G with at most $c|G|$ edges, where c is any constant, contain a set of 100 independent vertices?
10. Show that deleting at most $(m-s)(n-t)/s$ edges from a $K_{m,n}$ will never destroy all its $K_{s,t}$ subgraphs.
11. For $0 < s \leq t \leq n$ let $z(n, s, t)$ denote the maximum number of edges in a bipartite graph whose partition sets both have size n , and which does not contain a $K_{s,t}$. Show that $2\text{ex}(n, K_{s,t}) \leq z(n, s, t) \leq \text{ex}(2n, K_{s,t})$.
- 12.⁺ Let $1 \leq r \leq n$ be integers. Let G be a bipartite graph with bipartition $\{A, B\}$, where $|A| = |B| = n$, and assume that $K_{r,r} \not\subseteq G$. Show that

$$\sum_{x \in A} \binom{d(x)}{r} \leq (r-1) \binom{n}{r}.$$

Using the previous exercise, deduce that $\text{ex}(n, K_{r,r}) \leq cn^{2-1/r}$ for some constant c depending only on r .